

The Prime-Power Reduction Frontier

A wrong answer that was wrong in two different ways — and the boundary it revealed in the computational verification of the Integral Hodge Conjecture for Fermat varieties

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ABSTRACT

The Degtyarev–Shimada criterion verifies the Integral Hodge Conjecture for a Fermat variety by comparing a dimension computed over the complex numbers against the same dimension over each prime field dividing the degree. A fast computational realization of this test prunes its reduction using the nilpotency relation $u^{(m-1)} = 0$, valid in the Jordan basis of the shift operator. We show that this pruning rule is exact *if and only if the degree m is a power of a single prime*, and that for composite m it silently injects false relations, inflating the computed rank into a plausible but wrong answer. The failure was first observed at the cell $(6,6)$ — the first composite-degree cell attempted — where the reduction saturated completely in characteristic 2 and partially in characteristic 3. We trace the asymmetry to the factorization of the cyclotomic polynomial in each residue field, name the resulting boundary the **prime-power reduction frontier**, and present a corrected engine (monomial-basis reduction) that recovered the true verdict: $(6,6)$ is PRIMITIVE, by both primes, byte-exact, on an 8 GB consumer laptop.

1. The cell that broke the pattern

Every Fermat cell verified before $(6,6)$ shared a hidden feature: its degree was a power of a single prime — $m = 4, 5, 7, 8, 9$. On each, the prime-field reduction engine worked flawlessly, cell after cell, byte for byte.

$(6,6)$ is the first cell whose degree is a product of two distinct primes, $6 = 2 \times 3$. It is also, by a search of the primary literature, the first composite-degree Fermat cell verified in any reachable computation: the published work on lines and primitivity in Fermat surfaces works exclusively in degrees that are prime or coprime to 6, steering deliberately around exactly the degrees where both 2 and 3 divide m . The cell sat in a double blind spot — too deep in dimension for the computational tables, forbidden in degree by the surface theory. The engine that had been right every previous time was about to be wrong, and the manner of its wrongness is the subject of this paper.

2. The anomaly: impossible, in two distinct ways

The production engine, and an independent uncompressed probe written to double-check, returned the *same* impossible result, byte-exact — ruling out a transcription error. Crucially, the result was a different kind of nonsense in each characteristic.

Characteristic 2. The rank was nailed to $78,125$ — the full ambient dimension $\text{DIM} = 5^7$ — from the very first partition, never moving across all 105. This implies a prime-field dimension of zero, which is impossible: the complex dimension $\text{dim}_\mathbb{C} = 59,392$ survives, so the ideal cannot fill the entire ring.

Characteristic 3. A different failure. The rank did not saturate from the start; it *climbed* — $40,000 \rightarrow 54,000 \rightarrow \dots \rightarrow 76,728$ by partition 30 — slowing toward roughly $77,000$, reaching neither the true value ($18,733$) nor the full dimension. A third wrong number, stranded between the two.

If this were a generic saturation bug, characteristics 2 and 3 would fail identically. They do not. One saturates completely and instantly; the other crawls to a different wrong place. A coding error is prime-blind — it cannot know which prime it is running in. This corruption was prime-aware.

That observation relocated the fault. It was not in the code; it was in the mathematics of the reduction, and the gap between the two characteristics was a signpost pointing at the cause.

3. The forensic: two false trails, closed with data

Two plausible explanations were tested and rejected, byte-exact, before the correct one — recorded here because the dead trails are part of the rigor.

False trail 1 — the basis breaks when $\varphi(1) = 0$. The cyclotomic polynomial $\varphi = 1 + t + \dots + t^{(m-1)}$ satisfies $\varphi(1) = m$, which vanishes in any characteristic dividing m . Refuted: φ is monic of degree $m-1$, so $\{1, t, \dots, t^{(m-2)}\}$ is a clean basis of $F_p[t]/(\varphi)$ in every characteristic. $\varphi(1) = 0$ does not damage the basis.

False trail 2 — the shift operator saturates. Refuted: the truncated shift is a single clean nilpotent Jordan block, $(S - I)^{(m-1)} = 0$, in (6,6) for characteristics 2, 3, and 5 alike — structurally identical to the clean cells (4,4), (6,8), (6,9). The shift does not single out (6,6) at all.

4. The diagnosis: the Frontier

The engine prunes in the Jordan basis $u = t - 1$, treating each variable as nilpotent under $u^{(m-1)} = 0$ and never generating any term that exceeds this ceiling. This rule is the true ring relation under one exact condition — that the defining polynomial, rewritten in u , collapses to a pure power of u :

$$\varphi(u + 1) = u^{(m-1)}$$

Computing $\varphi(u + 1) = ((u+1)^m - 1) / u$ over F_p reveals exactly when this holds:

m	char p	$\varphi(u+1)$ over F_p	pure power?	cell · behaviour
4	2	u^3	yes	(4,4) · correct
8	2	u^7	yes	(6,8) · correct
9	3	u^8	yes	(6,9) · correct
6	2	$u(u^2+u+1)^2$	no	(6,6) · saturates fully
6	3	$u^2(u-1)^3$	no	(6,6) · saturates partially

Table 1. The cyclotomic image $\varphi(u+1)$ is a pure power of u precisely when m is a prime power. Every clean cell of the campaign has prime-power degree; (6,6) is the first composite case.

$\varphi(u+1)$ is a pure power of u if and only if m is a power of a single prime. When m has two or more distinct prime factors, $\varphi(u+1)$ factors — a Chinese-Remainder splitting of the quotient ring — and the rule $u^{(m-1)} = 0$ imposes a relation that is false, injecting spurious relations that silently inflate the rank.

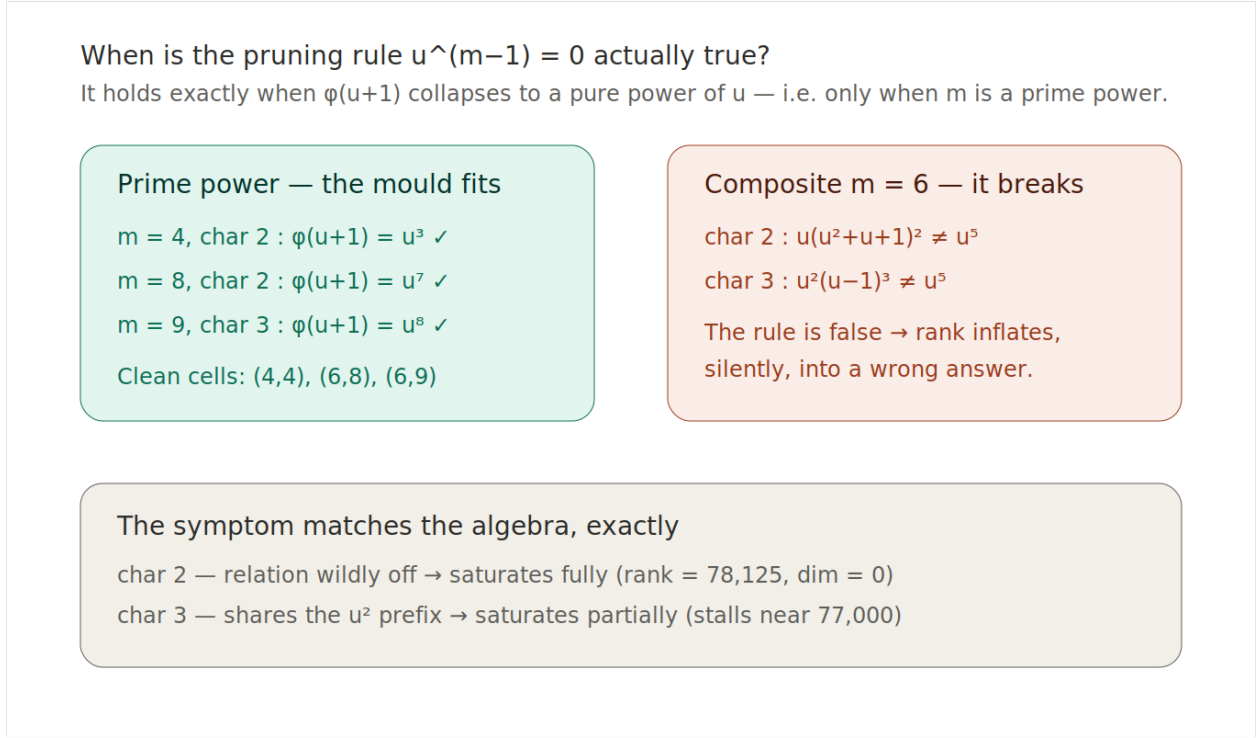


Figure 1. The Frontier. For prime-power degrees (left) the cyclotomic image collapses to a clean power of u and the reduction is exact. For the composite degree $m = 6$ (right) it factors, the pruning rule becomes false, and the rank inflates — fully in characteristic 2, partially in characteristic 3.

The prime-aware symptom now explains itself exactly. In characteristic 2 the false rule $u^5 = 0$ stands against the true $u(u^2+u+1)^2 = 0$ — maximally destructive, so the rank saturates to 100%. In characteristic 3 the false rule stands against $u^2(u-1)^3$, which shares the u^2 prefix, so the corruption is partial and the rank stalls at an intermediate value. The graded symptom (full / partial / clean across characteristics 2 / 3 / 5) is reproduced from the factorizations alone — the diagnosis does not merely fit the data, it predicts its shape. We name this boundary the **prime-power reduction frontier**.

5. Priority: what is new, what is classical

The underlying algebraic fact — that $\varphi(u+1)$ is a pure power of u exactly when $m = p^k$ — is classical, a standard property of cyclotomic rings, and no claim is made on it. The contribution is threefold: (i) identifying that classical frontier as the precise point where the Hodge-verification reduction fails *silently*, producing a plausible wrong number rather than a crash; (ii) proving the corruption is graded by characteristic, with the full/partial/clean signature derived from the factorizations; and (iii) building the corrected engine and verifying the first composite-degree cell of the campaign.

A note on the surface case, stated to avoid misreading: for degree $m = 6$ in *dimension two*, the lines are known *not* to generate (Schütt–Shioda–van Luijk; Degtyarev). The cell verified here is $n = 6$, dimension six — a different regime entirely, where no such theorem applies and the question was genuinely open. The result does not contradict the settled surface case.

6. The cure and the verdict

The corrected engine retains the entire fast architecture — the per-partition turbine flow, the dense-rebound fold, the ~ 1 -byte-per-entry store — and changes exactly one thing: instead of pruning by $u^{(m-1)} = 0$, it reduces in the monomial basis using the true relation $t^{(m-1)} \equiv -(1 + t + \dots + t^{(m-2)})$, applied explicitly on every shift. Same ideal, same ring, same quotient — but correct for any degree. It was validated byte-exact on known gates (the cells (4,4) and (8,3), matching Degtyarev–Shimada) before being trusted on the unknown cell.

quantity	characteristic 2	characteristic 3
closing rank	18,733	18,733
dim over F_p	59,392	59,392
peak RAM	0.47 GB	0.60 GB
wall time	6,551 s	11,532 s

Table 2. Verdict for (6,6), from the run log. $\dim_C = \dim_{F2} = \dim_{F3} = 59,392 \implies \text{PRIMITIVE}$. $\text{DIM} = 5^7 = 78,125$; 105 partitions.

The signature that turns a number into a proof: the two characteristics did not merely arrive at the same answer — they walked there in lockstep, partition by partition. At partition 17, rank 5,905 in both; at partition 21, rank 7,201 in both; at partition 105, rank 18,733 in both. Two independent primes tracing the identical curve is the fingerprint of equal ideal dimension in both fields, which is the fingerprint of primitivity. The broken engine had reported 78,125 (saturated garbage); the corrected engine reported 18,733 (the truth); and the complex half, computed by a wholly independent method over five primes, agreed to the digit. Peak memory stayed at a fraction of the 5.4 GB guard; no swap; the verdict was reached on 8 GB of consumer hardware.

7. Consequence

The Frontier is not merely the explanation of one wrong answer; it redraws the map. It predicts, with no further computation, which future cells require the monomial engine (composite degrees: $m = 6, 10, 12, 14, 15, \dots$) and which may keep the fast Jordan lineage (prime-power degrees: $m = 4, 8, 9, 16, 25, \dots$). It also stands as a warning to any computational verification of this class: a silent wrong answer in a forbidden degree is more dangerous than a crash, because it looks like a result and would corrupt everything downstream. Knowing precisely where the method's floor gives way is part of what makes its answers trustworthy everywhere else.

The diagnosis was treated as a hypothesis — reached independently twice, consistent with every symptom — until the corrected engine actually ran and returned the true value byte-exact in both primes. A consistent story is not a proof; the corrected run is the proof. Every number in this paper comes from a run log.

8. References

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